

Phase 1: Week 1 Detailed Task Allocation Plan

Project: Dynamic Spatio-Temporal GNN for Real-Time Fraud Topology & AML

Target Framework: Oracle Cloud ARM VPS / Memgraph / PyTorch Geometric & ONNX

Role Assignment Matrix

Role	Core Focus	Primary Responsibility (Week 1)
Member 1: DevOps	Cloud & Environment	Provisioning OCI, ARM performance tuning, and Docker deployment.
Member 2: Database	Graph Infrastructure	Memgraph initialization, 16GB RAM allocation, and schema design.
Member 3: Data Pipeline	ETL & Processing	Dataset preprocessing, sequence mapping, and batch ingestion scripts.
Member 4: ML Infra	PyG Tooling & Setup	Setting up ARM64 ML dependencies and validating sub-graph samplers.

Daily Task Breakdown per Member

Member 1: DevOps & Infrastructure Engineer (35-40 Hours)

- **Day 1-2:** Spin up OCI Always Free Ampere A1 (4 ARM Cores, 24GB RAM). Install Ubuntu 22.04 LTS ARM64. Configure VCN security lists to lock down ports 7687 and 3000.
- **Day 3:** Tweak system configurations via `/etc/security/limits.conf` (raise max open files to 65535). Disable transparent huge pages (THP) to fix database memory latency.
- **Day 4:** Install native ARM64 Docker CE and Docker Compose. Verify network bridge metrics and sandbox execution health.
- **Day 5:** Establish the base GitHub repository layout and integrate fundamental GitHub Actions workflows for continuous linting.

Member 2: Database & Knowledge Engineer (35-40 Hours)

- **Day 1-2:** Deploy the `memgraph/memgraph-platform:latest` container on the ARM instance. Modify `memgraph.conf` to dedicate a fixed 16GB RAM to the in-memory graph engine.
- **Day 3:** Finalise graph schema: `:Account (id, risk_score, is_licit)` and `:Transaction (id, amount, timestamp)`, bound by `:TRANSFERRED_TO` and `:OWNED_BY` edges.
- **Day 4:** Build database indexes on `Account(id)` and `Transaction(id)` to enforce $O(1)$ retrieval constraints. Establish a temporal index on `Transaction(timestamp)`.

- **Day 5:** Connect via Memgraph Lab. Execute mock Cypher statements and profile query execution logs to confirm index hits.

Member 3: Data Pipeline Engineer (35-40 Hours)

- **Day 1-2:** Ingest raw Elliptic and Kaggle Synthetic datasets. Design a python pandas alignment sequence to unify schemas.
- **Day 3:** Convert all database datetime records into clean, continuous UNIX epoch integers to facilitate time-window sampling steps.
- **Day 4:** Construct a chunked data ingestion wrapper script using the Bolt protocol. Target automated loading arrays in explicit blocks of 5,000 entities.
- **Day 5:** Fire a 10% structural stress pipeline run. Perform source-versus-destination validation to verify node and edge counts match completely.

Member 4: ML Infrastructure Engineer (35-40 Hours)

- **Day 1-2:** Build virtual development environments natively on ARM. Compile and verify PyTorch, torch-geometric, pyg-lib, and torch-scatter wheel binaries.
- **Day 3:** Design clean ingestion interfaces mapping Memgraph dynamic transaction states directly to PyG Data object configurations.
- **Day 4:** Prototype the dataloading frameworks using PyG NeighborLoader to safely capture k-hop graph neighborhoods without hitting storage limits.
- **Day 5:** Execute a mock structural forward pass through a basic GCN layer purely on the ARM CPU core arrays. Log latency and base RAM metrics.